that is,

$$s'_1 = \frac{n_g R s}{(n_g - 1)s - R} \quad . \qquad (3)$$

(2) Second image. Let s_2 be the object distance and s'_2 the image distance; then

$$\frac{1}{s'_2} + \frac{1}{s_2} = \frac{2}{r} \quad . \qquad (4)$$

Since $s_2 = 2R - s'_1$, $r = R$, (3), (4) give

$$s'_2 = \frac{(2s + 2R - n_g s)R}{3R + 3s - n_g s} \quad . \qquad (5)$$

(3) Third image. Let s_3 be the object distance and s'_3 the image distance; then

$$\frac{n_0}{s'_3} + \frac{n_g}{s_3} = \frac{n_0 - n_g}{r} \quad . \qquad (6)$$

Since $s_3 = 2R - s'_2$, $n_0 = 1$, $r = -R$, (5), (6) give

$$s'_3 = \frac{(4s - n_g s + 4R)R}{2n_g s - 4s + n_g R - 4R} \quad . \qquad (7)$$

2. Let v' be the image velocity; then

$$v' = \frac{\triangle s'_3}{\triangle t} = \frac{1}{\triangle t}\left\{\frac{[4(s+\triangle s) - n_g(s+\triangle s) + 4R]R}{2n_g(s+\triangle s) - 4(s+\triangle s) + n_g R - 4R} - \frac{(4s - n_g s + 4R)R}{2n_g s - 4s + n_g R - 4R}\right\} \qquad (8)$$

$$= \frac{-n_g^2 R^2 \triangle s / \triangle t}{(2n_g s - 4s + n_g R - 4R)^2 + \triangle s(2n_g - 4)(2n_g s - 4s + n_g R - 4R)}.$$

Since $\triangle s$ is small, terms containing $\triangle s$ in the denominator are neglected:

$$v' = \frac{-n_g^2 R^2}{(2n_g s - 4s + n_g R - 4R)^2} \frac{\triangle s}{\triangle t} \quad . \qquad (9)$$

As specified, P moves left to right at speed v; hence

$$v = -\frac{\triangle s}{\triangle t} \quad . \qquad (10)$$

Thus the image velocity is

$$v' = \frac{n_g^2 R^2 v}{(2n_g s - 4s + n_g R - 4R)^2} \quad . \qquad (11)$$

Image velocity depends on s; rectilinear motion is nonuniform. If

$$n = 2 \qquad (12)$$

, in (11) first two denominator terms cancel; v' is independent of s.

Motion is uniform, and (11) becomes $v' = v$, so the image and P have equal speeds.